THERMOCHEMISTRY

- Deals with heat changes caused by chemical Rx
- Based on first law of thermodynamics

Enthalpy of Rx: $\Delta H_{Rx} = \sum H_{Product} - \sum H_{Reactant}$

: ΔH_{Rx} is defined For Balanced Reaction

classification of ΔH_{rex}:

Endothermic Rx

- Heat is given to Rx
- $\Delta H_{Rex} > 0$
- $\Delta H_{Product} > \Delta H_{Reactant}$

Exothermic Rx

- Heat is evolved from Reaction
- $\Delta H_{rex} < 0$
- $\Delta H_{Product} < \Delta H_{Reactant}$

Factor affecting ΔH_{rex}:

① Physical state of substance:

$$2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(l)} \quad : \Delta H_1 = x$$
$$2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(g)} \quad : \Delta H_2 = y$$
$$\Big\} x \neq y$$

② Amount of substance:

$$2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(l)} \quad : \Delta H_1 = x$$
$$4H_{2(g)} + 2O_{2(g)} \longrightarrow 2H_2O_{(l)} \quad : \Delta H_2 = 2x$$

③ Allotropic form of substance:

$$C_{graphite} + O_{2(g)} \longrightarrow CO_{2(g)} \quad : \Delta H_1 = x$$
$$C_{diamond} + O_{2(g)} \longrightarrow CO_{2(g)} \quad : \Delta H_2 = y$$
$$\Big\{ x \neq y$$

④ Temperature of Rx:

Kirchoff's equation: $\dfrac{(\Delta H)_{T_2} - (\Delta H)_{T_1}}{T_2 - T_1} = \Delta n C_p$ at con. P

$$\frac{(\Delta V)_{T_2} - (\Delta V)_{T_1}}{T_2 - T_1} = \Delta n C_V$$

$\Big\{ \Delta n = n_{Product} - n_{Reactant}$

C_P, C_V = Heat capacity at const P, V

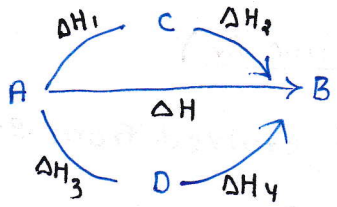
Laws of thermo-chemistry :

① Lavoiser Law : if R_x is reversed then ΔH_{rex} changes its sign

: $H_2 + \frac{1}{2} O_2 \longrightarrow H_2O$: $\Delta H = -285.8$ KJ

$H_2O \longrightarrow H_2 + \frac{1}{2} O_2$: $\Delta H = 285.8$ KJ

② Hess Law of constant Heat summation : ΔH_{rex} remains same whether it takes place in one step or more than one step



$\Delta H = \Delta H_1 + \Delta H_2$

$\Delta H = \Delta H_3 + \Delta H_4$

ΔH°_{rex} = standard enthalpy of reaction

standard condition : T = 25°C or 298 K

P = 1 atm)

Types of Enthalpy of Reaction

① Enthalpy of Formation ΔH°_f when 1mole of substance is formed from its constituent elements which are in their most stable form.

$H_{2(g)} + \frac{1}{2} O_{2(g)} \longrightarrow H_2O_{(\ell)}$: ΔH_f

$C_{graphite} + 2H_{2(g)} \longrightarrow CH_{4(g)}$: ΔH_f

$C_{Diamond} + O_{2(g)} \longrightarrow CH_{4(g)}$: $\Delta H \neq \Delta H_f$

. if conditions are not given than $\Delta_f H = \Delta_f H^{\circ} = (\Sigma \Delta_f H^{\circ})_{Product} - (\Sigma \Delta_f H^{\circ})_{Reactant}$

. $\Delta_f H^{\circ}$ in its most stable state assumed to be zero.

ex :- $\Delta_f H^{\circ}$ of $H_{2(g)}$, $N_{2(g)}$, $O_{2(g)}$, $Al_{(s)}$, $Cl_{2(g)}$, $Br_{2(\ell)}$, $C_{graphite}$, P_{white}, $S_{Rhomic} = 0$

② Enthalpy of combustion : $(\Delta_c H)$:

change in enthalpy when 1mole of a substance is completely burnt in Presence of $O_{2(g)}$

ex: $C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)}$: $\Delta_c H$: $\Delta_f H$

$C_{(s)} + \frac{1}{2} O_{2(g)} \longrightarrow CO_{(g)}$: $\Delta_c H$ ✗ $\Delta_f H$

- if condition are not given $\Delta_c H = \Delta_c H^\circ$

$$C_2H_{4(g)} + 3O_{2(g)} \longrightarrow 2CO_{2(g)} + 2H_2O_{(l)}$$

$$\Delta_c H^\circ_{C_2H_4(g)} = \Delta H^\circ_{reaction} = \left[2\Delta_f H^\circ_{CO_2} + 2\Delta H^\circ_{f\,H_2O}\right] - \left[\Delta_f H^\circ_{C_2H_4} + 3\Delta_f H^\circ_{O_2}\right]$$

- it is exothermic process : $\Delta H^\circ_{Rex} = \Delta H^\circ_{reactant} - \Delta H^\circ_{product}$

calorific value :- Heat release when 1mole of a substance completely burn in presence of excess of oxygen.

$$C_V = \frac{\Delta_c H}{\text{molecular weight}}$$

③ Enthalpy of fusion: ΔH°_{fusion} : Amount required to convert 1mole solid into Liquid at their melting point.

$$H_2O_{(s)} \longrightarrow H_2O_{(l)} \quad : \Delta H_{fusion} = +6\ KJ/mole$$

④ Enthalpy of vapourisation: $(\Delta_{vap}H^\circ)$:- Amount of heat required to convert 1mole liquid into gas at their Boiling point

$$H_2O_{(l)} \longrightarrow H_2O_{(g)} \quad : \Delta H_{vap} = +40.79\ KJ/mol$$

⑤ Enthalpy of sublimation: Amount of heat required to convert one mole solid into gas below their melting point

$$I_{2(s)} \longrightarrow I_{2(g)} \quad : \Delta H_{sub} = \Delta_{fusion}H + \Delta_{vap}H$$

⑥ Bond Enthalpy $(\Delta_{Bond}H)$: Amount of heat required to dissociate 1mole gas Bond into its gaseous state

ex:- For Diatomic molecules: $H\text{-}H_{(g)} \longrightarrow H_{(g)} + H_{(g)}$: $\Delta H = \Delta_{Bond}H$

$H\text{-}Cl_{(g)} \longrightarrow H^+_{(g)} + Cl^-_{(g)}$: $\Delta H \neq \Delta_{Bond}H$

$CH_4 \longrightarrow C_{(g)} + 4H_{(g)}$: $\Delta H = 400$ KJ/mol

: Average $\Delta_{C-H}H = \frac{400}{4} = 100$ KJ/mol

- In Polyatomic molecules Bond energy is average value. but in diatomic molecules Bond energy is absolute value

$CH_{4(g)} \longrightarrow CH_{3_g} + H_g$: $\Delta_{C-H} = 104$

$CH_{3(g)} \longrightarrow CH_{2_g} + H_g$: $\Delta_{C-H} = 98$

$CH_{2(g)} \longrightarrow CH_g + H_g$: $\Delta_{CH} = 99$

$CH_{(g)} \longrightarrow C_{(g)} + H_g$: $\Delta_{C-H} = 101$

Average $\Delta_{CH}H = 100$ K/cal/mol

⑦ Enthalpy of atomisation :-

Amount of heat required for the formation of one mole gaseous atom.

$H-H_{(g)} \longrightarrow 2H_{(g)}$: $\Delta_{at}H = \Delta H$

$H_2O_{(l)} \longrightarrow 2H_{(g)} + O_{(g)}$: $\Delta_{atm}H = \Delta H$

⑧ Enthalpy of Transtition : ($\Delta_{trs}H$) :

Enthalpy change when 1mole of an element converted into the one allotropic form to other allotropic form.

$C_{graphite} \longrightarrow C_{diamond}$: $\Delta_{trans}H = 1.90$ KJ/mole

⑨ Enthalpy of Hydrogenation :

Energy release when one mole Unsaturated compound converted into saturated compound by addition of Hydrogens.

$C=C + H_2 \longrightarrow C-C$: $\Delta H = \Delta H_{Hydrogenation}$

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(10) Enthalpy of Hydration : $\Delta H_{Hydration}$

. Energy release when 1 mole of anhydrous compound converted into Hydratid compound.

$$CuSO_{4(s)} + 5H_2O \longrightarrow CuSO_4 \cdot 5H_2O \quad : \Delta H = \Delta H_{Hydration}$$

· it is exothermic process because co-ordinate Bond formed between anhydrous compound and water

(11) Enthalpy of solution :

Enthalpy change when 1 mole of compound is completely dissolve in solvent so that further dissolution does not exist.

$$KCl_{(s)} + H_2O \longrightarrow KCl_{aq} \quad : \Delta H = \Delta H_{solution}$$

(12) Heat of Neutralisation ($\Delta H_{Neutral.}$) :

Energy changes when 1 gm-eq of an acid or Base is completely neutralised

case-1 : Neutralisation of strong Acid with strong Base

$$HCl + NaOH \longrightarrow NaCl + H_2O \quad : \Delta H_{neutrali} = -13.7 \text{ K cal.}$$

1 gm.eq 1 gm.eq

$$H^+ + Cl^- + Na^+ + OH^- \longrightarrow Na^+ + Cl^- + H_2O$$

· Heat of Neutralization of all SA & SB $= -13.7 \frac{\text{Kcal}}{\text{eq}} = -57.4 \text{ KJ/eq.}$

· Heat release when 1mole water is produce from H^+ & OH^- ion is called heat of neutralisation

· $H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O \quad : \Delta H_{neut} = 2 \times (-13.7) = -27.4 \text{ KCal}$

2gmeq 2gmeq

· Heat of Neutralisation = -57.4 KJ X No. of gm-eq of limiting Reagent

case-2 : Neutralisation of weak acid or weak Base with strong Acid or Base

$\Delta H_{neut.}$ is less than $57.4 \frac{KJ}{eq}$ or $13.7 \frac{K\,cal}{eq}$ because some of Heat used in the ionisation of weak acids or weak Base.

$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ $\quad \Delta H < 13.7$ K cal/eq.

$NH_4OH + HCl \longrightarrow NH_4Cl + H_2O$ $\quad \Delta H < 13.7$ K cal/eq

case-3 : Neutralisation of weak acid or Base with weak base or Acid.

$\Delta H_{neut} = -57.4$ KJ + Heat of ionisation of acid + Heat of ionisation of Base

$CH_3COOH + NH_4OH \longrightarrow CH_3COONH_4 + H_2O$: $\Delta H_{neut} = 50.$ KJ/eq

(13) Resonance Energy (R.E)

Difference in energy of most stable resonating structure and actual resonance Hybrid

R.E = observed Heat of formation − calculated Heat of Formation

(14) Lattice Energy ($\Delta_{Lattice}H$):

Energy release when 1mole ionic solid is prepared from their componet gaseous cation and anion.

$Na^+_g + Cl^-_g \longrightarrow NaCl_{(s)}$: $\Delta H_{Lattice} = (-)$ ve

$Na_{(s)} + \frac{1}{2} Cl_{2(g)} \longrightarrow NaCl_{(s)}$

$\downarrow \Delta H_{sub.}$ $\quad \downarrow BE/2$

$Na_{(g)}$ $\quad Cl_{(g)}$

$\downarrow I.E$ $\quad \downarrow E.A.$

$Na^+_{(g)}$ $\quad Cl^-_{(g)} \longrightarrow NaCl_s + U_{L.E}$

$\Delta H_{Rx} = \Delta H_{sub} + I.E + \frac{BE}{2} - EA - U$

$U \leftarrow$ Lattice energy.